Graph Sketching: Rectangular Hyperbola

Questions 1–6

Question 1. Sketch each of the following curves, in each case state asymptotes and any lines of symmetry:

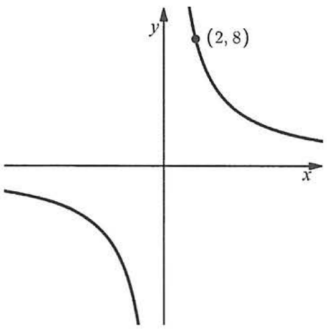
a) $xy = 9$
b) $xy = \frac{1}{4}$
c) $xy = -3$

Question 2. Each of the following curves are of the form

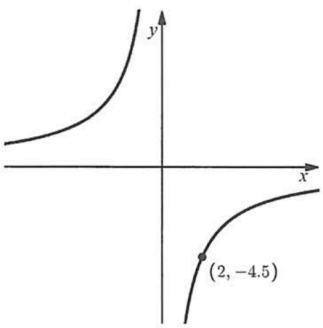
$$xy = a$$

where a is a constant. In each case state the value of a

a)



b)



Question 3. Show that the two asymptotes of the curve

$$x^2 - y^2 = k$$

are perpendicular to each other.

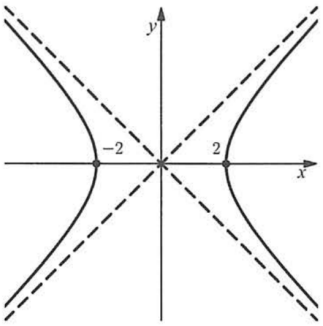
Question 4. Sketch each of the following curves, label each intersection with the axes clearly and include any asymptotes in your sketch.

a) $x^2 - y^2 = 1$
b) $x^2 - y^2 = 9$
c) $\dfrac{x^2}{2} - \dfrac{y^2}{2} = 1$
d) $y^2 - x^2 = 3$

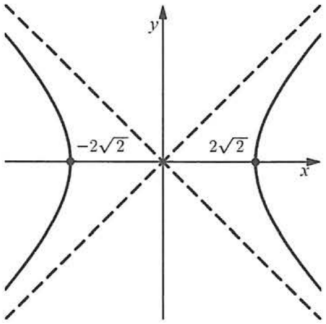
Question 5. For curves below find the value of the constant k, given each may be written in the form

$$x^2 - y^2 = k$$

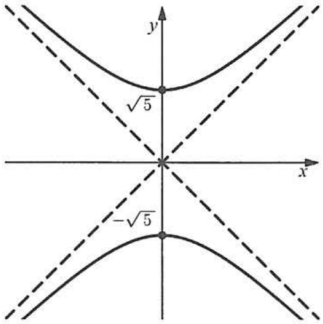
a)



b)



c)



Question 6. For constant k, a curve given by the equation

$$x^2 - y^2 = k$$

The curve passes through $(a, -\sqrt{b})$ for positive constants, a and b

a) Find the value of k in terms of a and b

b) Find the values at which it passes through the x-axis in terms of a and b